université de BORDEAUX

Automatic tuning of the IO500 benchmark parameters with Bayesian Optimization

M2 Computer Science
Internship Report

Inria Centre at the University of Bordeaux

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Supervisors: **[TODO:** supervisor names**]**

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1. Introduction

- **Context: I/O performance and the IO500 benchmark.**
  - HPC applications are increasingly limited by I/O rather than by compute: data volumes grow faster than the storage system's ability to absorb them.
  - Parallel file systems (Lustre, BeeGFS, GPFS) expose many tunable knobs (striping, number of storage targets, transfer size, process placement), and the right values are machine dependent.
  - IO500 is the de-facto standard benchmark for ranking HPC storage systems: it combines bandwidth-oriented and metadata-oriented phases into a single score, and publishes a biannual list.
  - Because IO500 is a *ranking*, sites have a direct incentive to tune it; the published score of a system therefore depends as much on how it was configured for the run as on the hardware itself.
- **Problem: proper tuning is out of reach.**
  - The IO500 submission rules require every write/create phase to run for at least 300 s (stonewall timer set to 300 s); a run whose write phase finishes earlier is marked **invalid**.
  - Consequence: one *valid* IO500 measurement costs roughly 30 min of exclusive cluster time.
  - The parameter space explored in this work has **[TODO:** confirm final number**]** $\approx$ 7,840 configurations; measuring all of them validly with 3 repetitions would take $\approx$ 12 000 h of wall-clock time, i.e. more than one year on the target machine.
  - Exhaustive tuning is therefore impossible in practice, and practitioners fall back on expert intuition or on greedy, one-parameter-at-a-time sweeps.
  - **Research question:** can a smarter search reach a near-optimal IO500 configuration in far fewer evaluations, and which search method actually does so?
- **Objectives of the internship.**
  - Build a *cheap* but faithful ground truth of the IO500 response surface on a real cluster.
  - Evaluate Bayesian Optimization (BO) as a tuning strategy for IO500, and determine *which* BO configuration works, since "Bayesian Optimization" is not a single method but a family (surrogate model $\times$ acquisition function $\times$ exploration parameter $\times$ budget).
  - Compare BO against random search, exhaustive search, and a human specialist.
  - Package the resulting methodology and configurations inside the IOPS framework, so that the study is reproducible by third parties.
- **Contributions.**
  - *A cheap ground-truth methodology.* Running the sweep with a shortened stonewall (10 s) and a small dataset produces *invalid* IO500 scores that nevertheless preserve the relative

ranking of configurations. The full grid was measured in $\approx 183\,\text{h}$ instead of more than a year.

  - *A reusable cache.* Every `parameters` $\rightarrow$ `SCORE` pair is stored in an SQLite database (`io500.db`), so any search strategy can be replayed offline at zero cluster cost with `iops run -cache-only`.
  - *A systematic comparison of BO setups*: **[TODO:** confirm**]** 83 BO configurations (surrogate $\in$ {GP, RF, ET}, acquisition $\in$ {EI, PI, LCB}, exploration $\xi/\kappa/n_{\text{init}}$, budget $\in \{10, 15, 20, 30, 40\}$), each repeated over 10 seeds.
  - *A sensitivity study* isolating which BO knob actually drives quality.
  - *A validation campaign* re-running the best configurations with a *valid* IO500 setup on the cluster, to check that the cheap proxy really does preserve the ranking.
  - *A human baseline*: a specialist challenge in which an expert submits their best guesses through a form, run under exactly the same conditions as the automated search.

- **Organisation of the report.**
  - **Context** presents the host laboratory, the team and the internship framework.
  - **Background** introduces IO500, parallel file system tuning and Bayesian Optimization.
  - **Related work** reviews existing approaches to I/O auto-tuning.
  - **Tools and experimental setup** describes the tools, the target platform and the complete installation procedure.
  - **Methodology** details how the ground truth was built and how the search strategies were compared.
  - **Results** reports and analyses the measurements.
  - **Discussion** covers the limitations and threats to validity, and **Conclusion** summarises the work and its perspectives.

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